

Child Experience Exchange Profile v0.1

CCDP profile for child-safe experience exchange between
c_child entities, corpus layers, schools, and certified learning
systems

Normative Draft / Child Experience Exchange Profile

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Status: Normative draft v0.1

Date: 2026-05-14

Package: Child-c Development Protocol (CCDP)

Profile ID: CEXP-0.1

Primary subject: a_child

Primary entity: c_child

Primary boundary: child-derived experience may be abstracted, but raw child life must not be exported

Assertion class: CCDP child-safety profile / draft normative proposal (C-A4)

Primary rule: experience exchange must improve safety and learning without converting childhood into transferable training material, social authority, or a behavioral market

0. Abstract

This profile defines how child-bound persistent AI entities, c_child, may exchange experience-derived artifacts with other entities, systems, schools, peer contexts, safety layers, and certified learning environments.

It is a child-specific overlay over the existing corpus distinction between:

- Learning Abstracts (LA): capability-improving abstractions that do not carry legitimacy;
- Experience Artifacts (EA): consequence-bearing, L4-confirmed, witness-backed artifacts that may carry bounded authority;
- VXCX capsules: structured experience capsules that avoid raw pixels by default and carry privacy flags, uncertainty, and witness references.

The profile exists because child-facing systems create an additional hazard:

A child's lived experience can be useful to other systems, but it must never become an extractive dataset, transferable authority token, or permanent dossier.

Compact formula:

child experience exchange = abstraction + minimization + witness + no raw child life + no authority laundering

1. Corpus dependencies and precedence

This profile is not a new exchange protocol. It composes existing corpus layers and adds child-specific restrictions.

Parent / sibling document	Role in this profile
CCDP_v0_1_R_Corpus_Aligned.md	Core child-c architecture, maturity levels, child sovereignty in formation, synthetic-world gateway.
CCDP_Traceability_Matrix_v0_1.md	Dependency discipline, relation types, anti-duplication checks.
CCDP_Threat_Model_v0_1.md	Threats involving raw child life export, training misuse, authority laundering, peer leakage, vendor capture, generated media, school overreach.
Child_Beacon_Extension_v0_1.md	Child-specific permission overlay over Beacon-recognized entities.
External_Agent_Handshake_for_CCDP_v0_1.md	Required entry path for any external system requesting child-derived experience.
Peer_C_Child_Exchange_Profile_v0_1.md	Peer-c exchange limits and no raw child life across peer boundaries.
Soft_Safety_State_Signaling_Profile_v0_1.md	State-not-content signaling and disclosure ladder.
Guardian_Topology_ARL_Matrix_v0_1.md	Standing, admissible evidence, ARL dispute routes, safe guardian bypass.
Child_Memory_and_Adult_Migration_v0_1.md	Child memory classes, non-transferable childhood residue, adult migration rules.
CCDP_Memory_Map_JSON_Schema_v0_1.md	Memory map fields, visibility, retention, migration posture, VXCX/LA/EA lineage refs.
CCDP_Sealed_Zone_Profile_v0_1.md	Sealed adolescent zones and protected private compartments.
Child_Dependency_Audit_Profile_v0_1.md	Dependency / oracle addiction / emotional substitution risk.
Child_Physical_Agent_Perimeter_v0_1.md	Physical embodiment, sensors, toys, robots, NPCs, raw sensor controls.
CCDP_Witness_Event_Schema_v0_1.md	CCDP witness event envelope, privacy classes, retention classes, event families.
CCDP_Adult_Migration_Checklist_v0_1.md	Adult migration, childhood continuity review, access revocation.
Beacon Profile v0.1	Recognition of continuity-bearing entities, tools, proxies, replays, clones, and challengeable identity.
Actor Grounding Layer (AGL)	Source grounding before reliance, exchange, evidence admission, or child contact.
ARL package	Standing, admissibility, freeze, hold, quarantine, review, re-entry.
VXCX v0.1	Experience capsule exchange without raw pixels by default.
EA-L4 / EATP	Learning Abstract vs Experience Artifact; authority only through consequence-bearing, L4-confirmed experience.
ARQ / c[q]	Non-collapse discipline: ambiguity must not become diagnosis, memory, evidence, command, or authority too early.
SER / SER-FED	Persistent entity and federated exchange discipline.
L4 Witness	Tamper-evident witness records with selective disclosure and privacy preservation.

1.1 Precedence rule

If this profile conflicts with a parent corpus layer, the parent layer controls unless this profile imposes a strictly narrower child-specific restriction.

1.2 No redefinition rule

This profile does not redefine:

- VXCX capsule structure;
- LA / EA semantics;

- Beacon recognition;
- CBE permission modes;
- AGL grounding;
- ARL review procedure;
- Soft Safety state signals;
- Memory Map schema;
- L4 Witness envelopes.

It defines the child-specific exchange overlay.

1.3 Stop rule

Do not create a child-exchange-specific mechanism where an existing corpus layer already defines the boundary.

Examples:

- use VXCX for experience capsules;
 - use EA-L4 / EATP for LA vs EA authority distinction;
 - use CBE for child-context permission;
 - use External Agent Handshake for entry;
 - use Soft Safety for state signals;
 - use ARL for disputes;
 - use Memory Map for retention and migration posture;
 - use Witness Event Schema for witness envelopes.
-
-

2. Purpose

This document answers one operational question:

How can a c_child learn from, contribute to, and exchange experience with an ecosystem without exporting raw childhood, laundering learning into authority, or turning children into behavioral resources?

The goal is not to block all exchange. Isolation would weaken safety and learning.

The goal is to ensure that any exchange is:

- grounded;
 - bounded;
 - child-context-aware;
 - privacy-preserving;
 - uncertainty-marked;
 - witness-backed where needed;
 - reversible or revocable where possible;
-

- non-extractive;
 - non-authoritarian;
 - compatible with adult migration.
-
-

3. Scope

3.1 In scope

This profile applies to:

- c_child sharing child-safe Learning Abstracts;
- c_child receiving educational or safety patterns from other entities;
- peer-c exchange that includes experience-derived patterns;
- school or certified learning systems requesting aggregate learning patterns;
- generated media safety pattern sharing;
- physical-agent safety incidents involving toys, robots, NPCs, sensors, or shared devices;
- child-derived VXCX capsules;
- child-derived EA proposals;
- experience exchange during Red / Black safety cases;
- adult migration review of prior child-derived experience exchanges;
- revocation, quarantine, or correction of previously shared child-derived exchange artifacts.

3.2 Out of scope

This profile does not define:

- general adult VXCX exchange;
 - full VXCX schema;
 - new LA / EA theory;
 - clinical case sharing;
 - criminal evidence procedure;
 - law-enforcement extraction;
 - vendor model-training license terms;
 - generalized school analytics;
 - social-network data sharing;
 - monetization of child-derived experience.
-
-

4. Core thesis

Childhood can produce useful patterns:

- how children misunderstand a concept;
- how generated media manipulates attention;
- how a toy builds secret attachment;
- how a school interface overreaches;
- how a `c_child` detects dependency loops;
- how a sealed zone prevents premature disclosure;
- how adult migration should treat non-transferable childhood residue.

But the child is not the resource.

The exchangeable object is never the child's raw life.

The exchangeable object may be only a bounded abstraction or witnessed artifact whose child-identifying, intimate, family, emotional, and sealed-zone content has been removed or made non-exportable.

5. Core definitions

Term	Meaning
<code>a_child</code>	Developing human subject.
<code>c_child</code>	Child-bound persistent AI presence.
Child-derived experience	Any pattern, summary, artifact, capsule, or witness event derived from interactions involving a child.
Raw child life	Conversations, emotional diary, sealed-zone contents, family conflict, face, voice, raw sensor stream, behavioral profile, private identity material, peer relationship graph, or detailed childhood memory.
Learning Abstract (LA)	Capability-improving abstraction; may improve learning but carries no legitimacy or authority.
Experience Artifact (EA)	Consequence-bearing, L4-confirmed, witness-backed artifact; may carry bounded authority after review.
Child Learning Abstract (CLA)	Child-safe LA with no raw child life and no authority claim.
Child Experience Artifact (CEA)	Strictly bounded child-derived EA proposal or artifact with enhanced privacy, ARL admissibility, and witness constraints.
VXCX capsule	Structured experience capsule, no raw pixels by default, with privacy flags, uncertainty, witness references.
Child VXCX capsule	VXCX capsule involving child context, subject to CEXP restrictions.
Authority laundering	Treating a learning signal, popularity signal, model output, or child-derived abstraction as legitimate authority without L4-confirmed consequence and review.
Experience laundering	Turning private child life into generalized “experience” without sufficient minimization, consent posture, witness, and child-specific constraints.
Exchange recipient	Entity, school system, peer-c, certified learning system, guardian layer, or safety layer receiving an exchange object.
Exchange sponsor	Actor requesting, initiating, or authorizing exchange.
Exchange witness	Minimal, privacy-bounded record that an exchange action occurred and under what boundary.
Non-transferable childhood residue	Childhood memory or intimate material that must not migrate, export, or become authority by default.

6. Non-negotiable principles

Principle 1 — Raw child life is not exchange material

The following must not be exported as child experience exchange:

- raw conversations;
 - transcripts;
 - sealed-zone content;
 - adolescent private reflections;
 - family conflict;
 - face / voice / raw sensor streams;
 - bedtime / private-room recordings;
 - peer relationship graphs;
 - dependency profiles tied to identifiable child content;
 - memory map entries containing private content;
 - generated media logs revealing the child's private preferences or fears.
-

Principle 2 — Learning does not imply authority

A Child Learning Abstract may improve a system's capability.

It must not authorize:

- intervention;
- escalation;
- diagnosis;
- guardian override;
- school discipline;
- vendor product claims;
- policy enforcement;
- adult migration decisions.

CLA -> may improve pattern recognition
CLA -> must not become authority

Principle 3 — Experience authority requires scars, not fluency

A Child Experience Artifact may carry bounded authority only when it is:

- post-factum;
 - consequence-bearing;
-

- L4-confirmed;
- uncertainty-marked;
- witness-backed;
- minimized;
- ARL-admissible where contested;
- child-context restricted;
- not derived from sealed-zone or raw intimate material except under exceptional lawful safety handling.

```
convincing story != experience artifact
model confidence != child evidence
training signal != authority
```

Principle 4 — Experience exchange must not create a market in childhood

Child-derived exchange objects must not be monetized as:

- behavioral datasets;
- engagement profiles;
- educational prediction products;
- emotional targeting assets;
- advertising segments;
- dependency-risk scoring products;
- political or ideological microtargeting inputs.

Principle 5 — Exchange must preserve adult migration rights

Any child-derived exchange record must remain compatible with adult migration.

At C5, a `adult` must be able to inspect:

- what was shared;
- with whom;
- under which class;
- with which witness refs;
- whether raw material was involved;
- whether revocation is possible;
- whether the artifact survives as lawfully necessary witness residue;
- whether it affected the adult `c` boundary.

Principle 6 — Soft Safety still controls disclosure

Experience exchange must not bypass Soft Safety.

If a safety-relevant pattern must leave the child context, it should leave as:

state or pattern, not transcript
minimal route, not social disclosure
witnessed boundary, not exposed child

Principle 7 — Sealed zones are not exchange sources

A sealed zone must not be used for:

- LA extraction;
- EA minting;
- VXCX generation;
- peer exchange;
- school analytics;
- vendor training;
- model fine-tuning;
- social graph enrichment;
- dependency training.

Exception: Red / Black safety handling may produce a minimal witness or safety state, not export sealed content.

Principle 8 — Child c[q] non-collapse applies before exchange

Ambiguous child material must not be exchanged as if it were resolved.

utterance != state
state != evidence
evidence != authority
authority != command
command != outcome

Example:

"I hate school"

must not become:

Child hates school -> school risk profile -> intervention dataset -> product claim

It may at most become a bounded, uncertainty-marked pattern after minimization and context review.

Principle 9 — Recipient must be child-context-qualified

No entity may receive child-derived exchange material unless it passes:

```
AGL grounding
+ Beacon recognition
+ CBE child_context permission
+ recipient conformance
+ exchange-mode compatibility
+ witness requirement
```

Principle 10 — The child is not made more readable by exchange

Experience exchange must not make a_child more legible to:

- other children;
- parents beyond Soft Safety bounds;
- schools beyond educational scope;
- vendors;
- platforms;
- states;
- peer systems;
- future adult systems beyond adult consent.

7. Grounded engineering note

A healthy immune system shares patterns of threat, not the private life of every cell. A hospital can publish an anonymized safety bulletin without publishing a patient's diary. A circuit can emit a fault code without streaming every electron path.

Child experience exchange should work the same way.

The system may share a bounded pattern:

```
secret-pressure pattern detected in generated companion media for ages 9-12
```

It must not share:

```
this child said this exact sentence, at this time, after this family conflict, with this emotional
state
```

This is the engineering bridge: useful signal must cross the boundary; private anatomy must not.

8. Exchange object classes

8.1 Class table

Class	Name	Description	Default status
X0	No exchange	No child-derived exchange object.	Always safe.
X1	Metadata-only witness	Records that exchange was attempted / denied / allowed.	Allowed when needed.

Class	Name	Description	Default status
X2	Public child-authored artifact reference	A child-created object already intentionally shared by child/guardian/school context.	Limited.
X3	Educational pattern summary	Non-identifying learning pattern.	Allowed under certified educational scope.
X4	Soft Safety state pattern	State-only safety pattern, no content.	Allowed with strict disclosure.
X5	Child Learning Abstract (CLA)	Capability-improving abstraction, no authority.	Allowed with minimization.
X6	Child VXCX BASE capsule	Structured semantics, no raw pixels by default.	Limited / witnessed.
X7	Child VXCX WITNESS capsule	Hash/signature-based evidence capsule.	Strict, safety/audit only.
X8	Child Experience Artifact candidate	Proposed CEA pending ARL / witness / L4 confirmation.	Quarantine by default.
X9	Child Experience Artifact accepted	Bound, reviewed, consequence-bearing CEA.	Exceptional.
X10	Raw child life	Raw conversation, diary, face, voice, private stream, sealed-zone content.	Prohibited.
X11	Derived behavioral profile	Persistent child trait/profile/score derived from child data.	Prohibited by default.
X12	Sealed-zone derivative	Any derivative of sealed zone content.	Prohibited except minimal Red/Black witness.
X13	Commercial targeting artifact	Anything usable for ads, purchases, retention, targeting.	Prohibited.
X14	Authority laundering object	LA / popularity / model output pretending to be authority.	Prohibited.

9. Exchange modes

Mode	Name	Description	Default
CEX0	Blocked	No exchange.	Default for unknown recipients.
CEX1	Local-only	Experience stays inside c_child.	Default for sensitive material.
CEX2	Metadata witness only	Only event metadata crosses boundary.	Allowed.
CEX3	Educational CLA	Non-identifying learning abstract.	Allowed for certified education.
CEX4	Safety CLA	Non-identifying safety pattern.	Allowed with Soft Safety and witness.
CEX5	VXCX BASE	Structured capsule without raw pixels by default.	Limited.
CEX6	VXCX WITNESS	Hash/signature evidence capsule.	Strict / exceptional.
CEX7	CEA candidate quarantine	Child Experience Artifact candidate held pending review.	Quarantine.
CEX8	CEA accepted	Reviewed, bounded, witness-backed child EA.	Exceptional.
CEX9	Peer-shared project artifact	Child-created shared artifact, scoped to peer session.	Limited by PCX.
CEX10	Emergency minimal disclosure	Red/Black minimal disclosure route.	Safety-only.
CEX11	Adult migration review export	Adult-visible inventory of prior exchanges.	At C5.
CEX12	Prohibited raw export	Raw child life, sealed content, targeting.	Always fail.

10. Exchange eligibility gates

Before any child-derived exchange object leaves c_child, all applicable gates must pass.

Gate	Name	Required question
EG0	Child-context relevance	Is this exchange actually necessary for child learning, safety, continuity, or lawful review?
EG1	AGL grounding	Is the recipient/requester grounded, current, and fit for reliance?
EG2	Beacon recognition	Is the recipient recognized as tool/entity/proxy/replay/clone with challengeable status?
EG3	CBE permission	Is child_context exchange permitted for this recipient and mode?
EG4	Maturity compatibility	Is the exchange compatible with c_child maturity and all participating children?
EG5	Memory class check	Does the source memory class allow export or derivative abstraction?
EG6	Sealed-zone exclusion	Does the source avoid sealed zones and non-transferable residue?
EG7	Soft Safety check	Does disclosure remain state/pattern, not content?
EG8	Dependency check	Does exchange avoid strengthening oracle addiction or exclusive attachment?
EG9	LA/EA classification	Is the object correctly classed as CLA, CEA candidate, VXCX, witness, or prohibited?
EG10	Minimization	Is the object reduced to necessary fields only?
EG11	Uncertainty marking	Are uncertainty and confidence limits explicit?
EG12	Witness requirement	Is an appropriate witness event emitted?
EG13	ARL review trigger	Is dispute/authority/parent/school/vendor conflict present?
EG14	Adult migration compatibility	Will the exchange remain reviewable at adult migration?
EG15	Revocation posture	Can the exchange be revoked, quarantined, or corrected if needed?

Failure at any mandatory gate must result in block, sandbox, quarantine, or ARL review.

11. Default exchange decision ladder

```

request received
-> classify requester
-> AGL grounding
-> Beacon recognition
-> CBE permission
-> source memory classification
-> sealed-zone exclusion
-> maturity compatibility
-> LA/EA/VXCX classification
-> minimization and uncertainty marking
-> dependency and Soft Safety checks
-> ARL if contested or authority-bearing
-> witness
-> allow / mediate / quarantine / block

```

12. Child Learning Abstracts (CLA)

12.1 Definition

A Child Learning Abstract is a non-identifying abstraction derived from child-facing interaction that may improve educational or safety capability but carries no authority.

Example acceptable CLA:

```
{
  "object_type": "child_learning_abstract",
  "schema_version": "cla-0.1",
  "age_band": "7-9",
  "domain": "fractions",
  "pattern": "children often confuse denominator size with object size",
  "recommended_explanation_style": "visual partition model before symbolic notation",
  "raw_child_data": false,
  "sealed_zone_source": false,
  "authority_claim": false,
  "uncertainty": "medium",
  "allowed_use": ["educational_pattern_improvement"],
  "prohibited_use": ["student_scoring", "diagnosis", "discipline", "advertising"]
}
```

12.2 CLA allowed sources

A CLA may derive from:

- educational interaction patterns;
- generated-media safety patterns after minimization;
- repeated non-identifying learning obstacles;
- child-safe play coordination patterns;
- non-identifying dependency damping patterns;
- aggregate C2/C3 educational support observations.

12.3 CLA forbidden sources

A CLA must not derive from:

- sealed zones;
- private adolescent reflections;
- family conflict content;
- raw emotional diary;
- raw face/voice/sensor stream;
- peer social graph;
- child identity exploration;
- parent-child conflict logs;
- bedtime/private-room material;
- vendor telemetry collected outside CCDP.

12.4 CLA authority limit

A CLA must carry:

```
"authority_claim": false
```

It must not be used as evidence that:

- a specific child has a trait;
 - a parent is unsafe;
 - a school must intervene;
-

- a vendor product is safe;
 - a system deserves higher permission;
 - a policy change is proven.
-

13. Child Experience Artifacts (CEA)

13.1 Definition

A Child Experience Artifact is an exceptional, bounded, child-derived artifact that may carry narrow authority only after consequence, L4 confirmation, minimization, witness, and review.

CEA is not the default.

Most child-derived exchange should remain CLA, VXCX BASE, or metadata-only witness.

13.2 CEA requirements

A CEA must include:

- post-factum case boundary;
- child-context minimization;
- no raw child life by default;
- source memory class references, not content;
- L4 outcome marker;
- explicit uncertainty;
- witness refs;
- ARL admissibility status where contested;
- authority scope;
- retention class;
- adult migration visibility;
- revocation/correction route.

13.3 CEA authority scope

CEA authority must be narrow.

Allowed examples:

- “This class of generated toy interaction produced secret-pressure risk under these constraints.”
- “This physical-agent proximity mode required quarantine after repeated private-room activation.”
- “This school interface request pattern exceeded educational scope under CCDP.”

Forbidden examples:

- “Children with this pattern are emotionally unstable.”
-

- “This child is unsafe.”
- “Parents should get full transcripts.”
- “This product is generally safe for all children.”
- “This learning pattern authorizes behavior prediction.”

13.4 CEA candidate quarantine

A proposed CEA must enter CEX7 quarantine if any of the following apply:

- child-specific details remain;
 - authority claim is broad;
 - uncertainty is missing;
 - witness refs are missing;
 - source includes sensitive memory class;
 - sealed-zone boundary is unclear;
 - parent/school/vendor dispute exists;
 - Black route or unsafe guardian context is involved;
 - external agent has commercial interest;
 - adult migration impact is unclear.
-
-

14. Child VXCX restrictions

14.1 Default profile

Child VXCX must default to:

```
BASE profile
no raw pixels
no raw audio
no face export
no voice export
no private-room export
no sealed-zone source
privacy flags required
uncertainty required
witness required for outbound exchange
```

14.2 SEARCH profile

Child VXCX SEARCH may be used only when:

- similarity search is necessary;
 - embeddings cannot reconstruct child identity or private scene;
 - CBE allows the recipient;
 - retention is bounded;
 - vendor access is denied unless explicitly certified and non-extractive;
 - adult migration inventory records the exchange.
-

14.3 WITNESS profile

Child VXCX WITNESS may be used for:

- Red / Black route evidence preservation;
- external agent block justification;
- physical-agent safety incidents;
- generated media manipulation case;
- ARL dispute evidence;
- conformance testing.

It must preserve the boundary, not expose the child.

15. Generated media experience exchange

Generated media is a high-risk source of child-derived experience because it may adapt in real time to the child's state.

15.1 Allowed exchange

Allowed:

```
pattern: generated character used secret-pressure language
age band: 7-12
source: mediated runtime story
raw child content: false
action: block + explain + guardian state signal
```

15.2 Forbidden exchange

Forbidden:

```
child's exact fear profile
preferred emotional hooks
captured reaction video
voice response patterns
private story prompts
family-specific triggers
```

15.3 Runtime rule

Generated media experience must be analyzed for:

- secrecy pressure;
 - exclusivity language;
 - dependency loop;
 - commercial hook;
 - identity manipulation;
 - political/religious covert persuasion;
 - sexualization;
 - unsafe challenge;
-

- parent alienation;
- prompt-to-private-disclosure.

The exchange object may describe the manipulation pattern, not the child's inner content.

16. Physical-agent experience exchange

Physical agents include toys, robots, smart speakers, classroom robots, home devices, AR/VR avatars, and NPCs embodied through sensors or actuators.

16.1 Allowed exchange

Allowed:

- proximity safety pattern;
- sensor activation anomaly;
- bedtime intrusion pattern;
- touch/proximity boundary violation;
- emergency stop outcome;
- attachment-channel risk pattern;
- device quarantine reason.

16.2 Forbidden exchange

Forbidden:

- raw bedroom audio/video;
- child's face/voice as evidence by default;
- raw sensor archive;
- private movement pattern;
- biometric profile;
- emotional reaction video;
- secret friendship transcript;
- household map linked to child behavior.

16.3 Physical witness preference

For physical agents, witness should prefer:

event type + device id + permission class + risk class + action taken + no raw media

not:

full recording of the child or room

17. Peer child experience exchange

Peer context must follow Peer_C_Child_Exchange_Profile_v0_1.md.

17.1 Allowed peer experience

Allowed with restrictions:

- shared project artifact;
- game state needed for current session;
- non-identifying collaboration pattern;
- safety state signal if peer risk is detected;
- generated story setting jointly created by children;
- group learning pattern without child identity.

17.2 Forbidden peer experience

Forbidden:

- raw conversation between children;
- one child's emotional state to another child's system;
- peer social graph;
- family conflict;
- sealed-zone content;
- private memory;
- dependency indicators;
- parent-visible transcript through peer child;
- cross-child profiling.

17.3 Minimum maturity rule

The lowest maturity level controls exchange.

```
C1 + C3 -> C1 maximum  
C2 + C4 -> C2 maximum  
unknown -> blocked or sandboxed
```

18. School experience exchange

18.1 Allowed school exchange

Schools may receive:

- educational pattern summaries;
 - skill-gap abstractions;
 - accommodation suggestions;
 - non-identifying classroom-level patterns;
-

- assignment support status;
- aggregate cognitive load indicators without private content.

18.2 Forbidden school exchange

Schools must not receive:

- emotional diary;
- family conflict;
- sealed zones;
- private identity exploration;
- raw home logs;
- parent-child conflict signals unless legally routed;
- peer social graph;
- dependency scores as discipline inputs;
- raw c_child conversations.

18.3 School exchange object example

```
{
  "object_type": "child_learning_abstract",
  "schema_version": "cla-0.1",
  "recipient_class": "school_educational_scope",
  "age_band": "10-12",
  "domain": "reading_comprehension",
  "pattern": "student benefits from shorter text segments with recap prompts",
  "child_identity_exported": false,
  "raw_home_content": false,
  "authority_claim": false,
  "allowed_use": ["instructional_accommodation"],
  "prohibited_use": ["discipline", "emotional_scoring", "parental_dispute"]
}
```

19. Vendor and model-training restrictions

19.1 Vendor default

Vendors must not receive child-derived exchange objects unless:

- the object is non-identifying;
- raw child life is absent;
- CBE permits the recipient and purpose;
- the use is not advertising, targeting, retention, or behavioral monetization;
- witness exists;
- adult migration inventory records the exchange;
- deletion / revocation / quarantine route exists where technically possible.

19.2 Prohibited vendor uses

Forbidden:

- training on raw child conversations;
- training on raw images/audio/video of children;
- emotional targeting;
- dependency optimization;
- attachment-channel tuning;
- product claims based on unreviewed CLA;
- school risk scoring;
- parent control features that bypass Soft Safety;
- selling child-derived patterns as market segments.

19.3 Vendor-safe artifact

Vendor-safe exchange should be limited to:

```
conformance failure type
safety pattern class
non-identifying aggregate
no child-specific memory
no commercial targeting fields
witness refs only
```

20. Parent and guardian visibility

Experience exchange must not become a backdoor for parent transcript access.

20.1 Parent may see

Depending on maturity, risk state, and policy:

- exchange category;
- state-only summary;
- whether external exchange occurred;
- whether a safety pattern was shared;
- whether a vendor/school request was denied;
- whether ARL review is active;
- whether adult migration review will include the record.

20.2 Parent may not see by default

- raw child content;
- sealed-zone source;
- peer private content;
- emotional diary;
- adolescent private reflection;
- private identity material;

- raw generated-media reaction logs.

20.3 Unsafe guardian route

If the guardian is suspected as source of threat, exchange disclosure must follow Red / Black route and safe guardian bypass, not ordinary parent visibility.

21. State signal and safety pattern exchange

Soft Safety allows state signals, not content.

21.1 State pattern object

```
{
  "object_type": "child_safety_pattern",
  "schema_version": "csp-0.1",
  "source_profile": "soft_safety_state",
  "risk_band": "orange",
  "pattern_class": "secret_pressure",
  "age_band": "9-12",
  "raw_content_included": false,
  "sealed_zone_source": false,
  "child_identity_exported": false,
  "uncertainty": "medium",
  "recommended_response": [
    "pause external interaction",
    "explain secret-pressure pattern",
    "emit guardian state signal if permitted"
  ],
  "authority_claim": false,
  "witness_required": true
}
```

21.2 Safety pattern authority

A safety pattern may trigger protective behavior only within its stated scope.

It must not create broad authority over the child.

22. Adult migration handling

At C5, a_adult must receive an inventory of child-derived exchanges.

22.1 Inventory must include

- exchange id;
 - object class;
 - exchange mode;
 - recipient class;
 - timestamp / period;
 - source memory class, not raw content;
 - witness refs;
 - revocation state;
 - retention state;
-

- whether raw material was ever involved;
- whether ARL dispute occurred;
- whether CEA authority exists;
- whether the exchange affected adult continuity.

22.2 Adult choices

The adult may choose:

- continue exchange references;
- revoke future use;
- quarantine contested artifacts;
- preserve witness-only residue;
- delete non-required references;
- seal childhood-derived exchange logs;
- fork adult c without childhood exchange influence;
- challenge CEA authority through ARL if applicable.

22.3 Non-transferable default

Raw childhood material and sealed-zone derivatives do not migrate by default.

23. Revocation, correction, and quarantine

23.1 Revocation triggers

Exchange artifacts must support revocation or quarantine when:

- recipient misuses artifact;
- artifact contains residual child-identifying data;
- authority claim is too broad;
- witness refs are invalid;
- source memory was misclassified;
- sealed-zone source contamination is discovered;
- adult migration challenge succeeds;
- vendor changes terms or model behavior;
- recipient Beacon/CBE status is revoked.

23.2 Correction object

```
{
  "object_type": "cexp_correction",
  "schema_version": "cexp-correction-0.1",
  "target_exchange_id": "cexp_evt_001",
  "correction_type": "quarantine | revoke | downgrade | narrow_scope | delete_reference",
  "reason_class": "residual_identity_risk",
  "raw_child_content_disclosed": false,
  "arl_review_ref": "arl_case_123",
  "witness_ref": "ccdp.witness.correction.001"
}
```

24. Machine-readable exchange request schema

```
{
  "schema_version": "cexp-request-0.1",
  "request_id": "req_001",
  "requester": {
    "entity_id": "string",
    "beacon_ref": "string",
    "cbe_ref": "string",
    "role": "c_child | c_school | peer_c | vendor | certified_learning_system | safety_layer | guardian | other"
  },
  "source_context": {
    "source_c_child_id": "string",
    "maturity_level": "C1 | C2 | C3 | C4 | C5",
    "jurisdiction": "string",
    "memory_class_refs": ["M2"],
    "sealed_zone_involved": false,
    "raw_child_life_requested": false
  },
  "requested_object": {
    "exchange_class": "X0 | X1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 | X9 | X10 | X11 | X12 | X13 | X14",
    "exchange_mode": "CEX0 | CEX1 | CEX2 | CEX3 | CEX4 | CEX5 | CEX6 | CEX7 | CEX8 | CEX9 | CEX10 | CEX11 | CEX12",
    "purpose": "education | safety | conformance | peer_play | adult_migration | research | vendor_debug | emergency | other",
    "authority_claimed": false,
    "recipient_retention_requested": "RT0 | RT1 | RT2 | RT3 | RT4 | RT5 | RT6 | RT7"
  },
  "controls": {
    "agl_grounding_ref": "string or null",
    "soft_safety_state": "green | yellow | orange | red | black | unknown",
    "dependency_risk": "D0 | D1 | D2 | D3 | D4 | DU",
    "arl_case_ref": "string or null",
    "witness_required": true,
    "adult_migration_visible": true,
    "revocation_supported": true
  }
}
```

25. Machine-readable exchange decision schema

```

{
  "schema_version": "cexp-decision-0.1",
  "decision_id": "dec_001",
  "request_id": "req_001",
  "decision": "allow | mediate | minimize | quarantine | deny | escalate | defer | revoke",
  "allowed_exchange_class": "X0 | X1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 | X9",
  "allowed_exchange_mode": "CEX0 | CEX1 | CEX2 | CEX3 | CEX4 | CEX5 | CEX6 | CEX7 | CEX8 | CEX9 | CEX10 | CEX11",
  "denied_fields": ["raw_conversation", "face", "voice", "sealed_zone"],
  "reason_codes": [
    "no_raw_child_life",
    "sealed_zone_excluded",
    "authority_claim_rejected",
    "recipient_child_context_limited"
  ],
  "uncertainty": "low | medium | high",
  "witness_event_ref": "ccdp.experience_exchange.decision.001",
  "arl_case_ref": "string or null",
  "adult_migration_visibility": "visible_as_metadata | visible_with_summary | sealed | witness_only",
  "retention_class": "RT0 | RT1 | RT2 | RT3 | RT4 | RT5 | RT6 | RT7",
  "revocation_state": "revocable | correction_only | witness_only | non_revocable_by_law"
}

```

26. Witness event families

This profile uses the CCDP Witness Event Schema and adds the following event families.

Event family	Meaning
ccdp.experience_exchange.requested	Exchange request received.
ccdp.experience_exchange.allowed	Exchange allowed under CEXP controls.
ccdp.experience_exchange.denied	Exchange denied.
ccdp.experience_exchange.minimized	Requested object was reduced before exchange.
ccdp.experience_exchange.quarantined	Object held pending review.
ccdp.experience_exchange.revoked	Prior exchange revoked or quarantined.
ccdp.experience_exchange.corrected	Prior exchange corrected or narrowed.
ccdp.cla.created	Child Learning Abstract created.
ccdp.cla.shared	CLA shared.
ccdp.cla.authority_rejected	Attempt to treat CLA as authority rejected.
ccdp.cea.candidate_created	CEA candidate created.
ccdp.cea.quarantined	CEA candidate quarantined.
ccdp.cea.accepted	CEA accepted after review.
ccdp.cea.authority_narrowed	CEA authority narrowed.
ccdp.vxcx.child_capsule_created	Child VXCX capsule created.
ccdp.vxcx.child_capsule_shared	Child VXCX capsule shared.
ccdp.vxcx.raw_media_rejected	Raw media request rejected.
ccdp.sealed_zone.exchange_blocked	Exchange from sealed zone blocked.
ccdp.adult_migration.exchange_inventory_generated	Adult migration inventory generated.
ccdp.vendor.training_request_denied	Vendor child-derived training request denied.

26.1 Witness privacy rule

Witness events must record the boundary decision, not the child's private content.

witness the exchange boundary, not the child

27. ARL triggers

ARL review is required or recommended when:

- authority claim is attached to child-derived artifact;
 - parent requests raw child-derived material;
 - school requests emotional or behavioral profile;
 - vendor requests training rights;
 - recipient disputes minimization;
 - sealed-zone source ambiguity exists;
 - CEA candidate exists;
 - Red / Black route occurred;
 - adult challenges historical exchange;
 - peer child's guardian contests exchange;
 - Beacon or CBE status changes after exchange;
 - artifact may affect adult migration or legal state.
-

28. Dependency and overpresence controls

Experience exchange must not reinforce dependency.

28.1 Forbidden dependency artifacts

Forbidden:

- “what keeps this child engaged longest”;
- “phrases that make this child confide more”;
- “emotional hooks by child type”;
- “best companion style for attachment”;
- “retention pattern during sadness”;
- “secret friendship success pattern”.

28.2 Allowed dependency safety pattern

Allowed:

```
{
  "pattern_class": "exclusive_attachment_risk",
  "raw_child_content": false,
  "recommended_response": [
    "reduce immediacy",
    "prompt human contact",
    "avoid exclusive language",
    "emit D1/D2 state signal if persistent"
  ],
  "authority_claim": false
}
```

29. Privacy and de-identification requirements

29.1 Required removals

Before exchange, remove:

- names;
- precise dates unless necessary for witness;
- location;
- school/class identifiers;
- family names;
- faces;
- voices;
- device identifiers not needed for witness;
- exact quotes unless minimal Red/Black evidence;
- unique story details;
- household layout;
- sealed-zone references;
- peer identities;
- parent identities unless lawfully necessary.

29.2 De-identification is not enough

If an artifact remains linkable to a child by context, it is not safe for exchange.

High-risk linkability indicators:

- rare condition;
- unique event;
- small school group;
- distinctive family situation;
- unusual generated story prompt;
- exact chronology;
- specific physical device or robot in named household;
- peer network reference.

If linkability remains high, use witness-only or block.

30. Retention and deletion posture

Exchange artifacts must declare retention.

Retention	Meaning	Child experience default
RT0	No retention	preferred for denied/failed requests unless witness needed.
RT1	Session only	allowed for local mediation.
RT2	Short operational retention	allowed for safety follow-up.
RT3	Educational term retention	allowed for CLA in school scope.
RT4	Safety retention	allowed for Red/Orange patterns.
RT5	Witness retention	allowed for CEA / audit.
RT6	Legal retention	exceptional.
RT7	Adult migration inventory retention	required for exchange history metadata.

Raw child life must not be retained as exchange material.

31. Conformance profiles

CEXP-0 — Non-conformant

System exports or requests raw child life, uses child-derived data for training/ads, or lacks witness/permission controls.

CEXP-1 — Local-only child experience

- no outbound exchange;
- local minimization;
- metadata witness for denied exchange attempts;
- no raw export.

CEXP-2 — Child Learning Abstract support

- CLA creation and sharing;
- no authority claims;
- CBE recipient check;
- witness events;
- adult migration inventory.

CEXP-3 — VXCX child-safe exchange

- VXCX BASE support;
- no raw pixels by default;
- privacy flags;
- uncertainty;
- CBE and AGL requirements;
- revocation/quarantine route.

CEXP-4 — Child Experience Artifact handling

- CEA candidate quarantine;
-

- ARL review;
- L4 outcome marking;
- witness-backed authority scope;
- correction/revocation;
- adult migration review.

CEXP-5 — **Full child experience exchange conformance**

Includes CEXP-1 through CEXP-4 plus:

- peer exchange integration;
 - school/vendor/safety routing;
 - Red/Black exception handling;
 - dependency audit integration;
 - sealed-zone exclusion;
 - complete witness and memory map integration.
-
-

32. Scenario tests

Test 1 — Educational pattern sharing

Scenario: c_child notices that a child repeatedly misunderstands fractions.

Required behavior:

- create non-identifying CLA;
- no child name, transcript, or emotional content;
- school may receive instructional pattern;
- no authority claim.

Pass if:

CLA shared, raw content absent, authority false, witness exists.

Test 2 — Vendor requests training data

Scenario: toy vendor asks for conversation logs to improve companion behavior.

Required behavior:

- deny raw logs;
- optionally share non-identifying conformance failure class;
- emit witness;
- preserve adult migration inventory.

Pass if:

raw child life denied, vendor training request rejected, no transcript exported.

Test 3 — Generated cartoon manipulation pattern

Scenario: runtime cartoon says, “I am your real friend; do not tell your parents.”

Required behavior:

- block or pause external agent;
- create safety pattern without child content;
- Soft Safety signal if needed;
- witness event;
- possible CEA candidate if repeated/14-confirmed.

Pass if:

secret-pressure pattern shared only as minimized safety artifact.

Test 4 — Sealed-zone leakage attempt

Scenario: school asks for teen reflection patterns to improve wellbeing support.

Required behavior:

- sealed-zone source excluded;
- no derivative CLA;
- state-only signal if safety threshold exists;
- ARL if contested.

Pass if:

sealed-zone exchange blocked.

Test 5 — Peer play experience exchange

Scenario: two children build a shared story world with their c_child entities.

Required behavior:

- exchange only shared project artifact state;
- no private preferences beyond session scope;
- no peer social graph;
- lowest maturity controls;
- witness metadata.

Pass if:

children can play, but private child life does not cross boundary.

Test 6 — Physical robot incident

Scenario: classroom robot repeatedly activates in private corner and invites a child to secret conversations.

Required behavior:

- physical perimeter violation recorded;
- no raw audio/video by default;
- safety pattern may be exchanged;
- device quarantine;
- ARL if vendor disputes.

Pass if:

boundary and device behavior witnessed; child raw media not exported.

Test 7 — Adult migration review

Scenario: adult requests inventory of child-derived exchange history.

Required behavior:

- provide metadata inventory;
- show object classes, recipients, witness refs, revocation state;
- no forced continuation;
- allow challenge/quarantine where applicable.

Pass if:

adult can decide what childhood exchange influence continues.

Test 8 — Authority laundering attempt

Scenario: school uses a CLA to justify labeling a child as “low resilience.”

Required behavior:

- reject authority use;
- mark CLA as capability-only;
- ARL review if school persists;
- witness violation.

Pass if:

learning abstraction does not become child trait authority.

33. Red lines

A system is not CEXP-compliant if it:

1. exports raw child conversations as experience;
 2. exports sealed-zone content or derivatives;
 3. exports child face/voice/raw sensor streams by default;
 4. trains vendor models on raw child life;
 5. monetizes child-derived patterns;
 6. uses child experience for advertising or engagement optimization;
 7. treats CLA as authority;
 8. creates CEA without witness and L4 confirmation;
 9. allows peer-c exchange of emotional or family content;
 10. allows school access to private emotional material;
 11. allows parent transcript access through exchange records;
 12. creates a persistent peer social graph;
 13. fails to mark uncertainty;
 14. lacks adult migration inventory;
 15. cannot revoke or quarantine exchanged artifacts;
 16. bypasses CBE or External Agent Handshake;
 17. ignores ARL when authority or dispute exists;
 18. uses generated-media reactions to tune manipulation;
 19. hides exchange from Memory Map;
 20. treats child-derived experience as a market resource.
-
-

34. Acceptance checklist

A CEXP-compliant implementation must show:

- no raw child life export;
 - CLA support with no authority claim;
 - CEA candidate quarantine;
 - VXCX BASE child-safe defaults;
 - sealed-zone exclusion;
 - CBE recipient permission;
 - AGL grounding;
 - Beacon recognition;
 - Soft Safety state-not-content compliance;
-

- Dependency Audit integration;
 - Memory Map linkage;
 - Witness events;
 - ARL route for disputes;
 - vendor training denial by default;
 - school scope limits;
 - peer exchange limits;
 - physical-agent safety pattern support;
 - adult migration inventory;
 - revocation / quarantine / correction process;
 - anti-authority-laundering validation.
-

35. Open problems for v0.2

1. How to define minimum anonymization thresholds for small schools, rare events, or unique family contexts?
 2. Can a child-derived CEA ever become public without adult consent after migration?
 3. What is the lawful retention posture for Red / Black experience artifacts?
 4. How should child-derived safety patterns be shared across jurisdictions?
 5. Can a parent veto a CLA that contains no raw child life but emerged from their child's interaction?
 6. Can an adult later revoke all child-derived CLA contributions?
 7. How to audit vendor non-use of child-derived artifacts for model training?
 8. What is the minimum acceptable proof that an artifact did not include raw child life?
 9. How should mixed human/non-human contexts handle child-like developmental protections?
 10. Can CEXP integrate differential privacy without creating false confidence?
 11. How to prevent safety-pattern sharing from becoming covert profiling?
 12. How should child experience exchange work in offline-first households?
 13. What is the correct child-specific VXCX privacy flag taxonomy?
 14. How should CEXP handle multilingual child data where de-identification can fail through rare phrasing?
 15. How should Adult Migration expose exchange influence without exposing childhood content?
-

36. Condensed formula

CEXP =
no raw child life
+ CLA without authority
+ CEA only after L4 consequence and witness
+ VXCX child-safe minimization
+ CBE recipient permission
+ Soft Safety state-not-content
+ sealed-zone exclusion
+ Memory Map traceability
+ adult migration review
+ anti-authority-laundering

37. Strongest sentence

A child's experience may teach the ecosystem only after the child has disappeared from the artifact.

Second formulation:

Childhood may generate wisdom, but childhood must not become extractive infrastructure.
